

Samuele Pironaci

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ABOUT

Computer Science undergraduate with expertise in C++ and scalable Java/Node.js architectures. Specialist in designing robust REST APIs, managing diverse data layers (PostgreSQL, MongoDB), and bridging the gap between complex backend logic and intuitive UI/UX design.

EDUCATION

University of Calabria

Present

Bachelor Degree: Undergraduate in Computer Science and Artificial Intelligence

- **Main Courses:** Discrete Mathematics, Data Structures and Algorithms, Web Application, User Interface Design, Network and Security, Database, Operating Systems, Parallel Algorithms and Distributed Systems, Artificial Intelligence.

EXPERIENCE

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| • Part-time waiter at the Drop in Soverato | June-August 2022 |
| • Part-time Waiter/Cashier at Chocolat in Soverato | June-August 2023 |

TECHNICAL SKILLS

Languages: C/C++, Java, Rust(learning...), Javascript, Typescript, Python, SQL, HTML, CSS, ASP(Answer Set Programming).

Tools and Frameworks: MPI, POSIX Threads, Node.js, React, Angular, Git, GitHub.

Databases: MySQL, PostgreSQL, MongoDB.

DevOps: Docker, CI/CD, RESTful Design.

PROJECTS

AuraBrain Personal Project (In progress)

March 2026

- Development of a modern web application utilizing the MERN stack (MongoDB, Express, React, Node.js).
- Implementation of a REST API architecture with Node.js for seamless data flow management.
- Creation of a dynamic and responsive user interface using React and advanced state management.
- Data modeling on MongoDB non-relational database.
- Implementation of advanced semantic search features via MongoDB Atlas Vector Search, managing embeddings for intelligent data retrieval.
- Containerization with Docker.

Blog Ricettario (Ricette Facili)

January 2026

- Development of a decoupled architecture featuring a Spring Boot backend and an Angular frontend.
- Design and implementation of REST APIs for real-time data management.
- Integration of a PostgreSQL database for robust data persistence.
- UI/UX module management to ensure a responsive and intuitive user interface.

Food Delivery+

September 2025

- Development of a desktop/backend application based on the Model-View-Controller (MVC) design pattern, pattern DAO and Singleton.
- Implementation of business logic in Java with a strong focus on separation of concerns.
- Data management through JDBC and SQL, with the development of a functional user interface.
- Focused on code maintainability and the rigorous application of Object-Oriented Programming (OOP) principles.

Distributed Multi-Agent AI System – Tron Edition

- Developed a real-time strategic simulation where multiple AI agents compete in a grid-based environment. The core challenge was decoupling the game's physical state from the agents' decision-making process, achieved through a declarative logic layer.
- Declarative AI Engine: Integrated the DLV2 solver via the EmbASP framework to handle complex spatial reasoning. The AI evaluates thousands of possible outcomes per tick to ensure collision avoidance (Borders, Trails, and Peer-to-Peer).
- Robust Architectural Design: Implemented a strictly decoupled MVC (Model-View-Controller) pattern.
- Multi-Agent Coordination: Authored ASP logic to manage 4 simultaneous agents, ensuring global stability and preventing "U-turns" and mutual collisions through integrity constraints.

SOFT SKILL

- Pragmatic Problem Solving & Decision Making.
- Fast Learner & Continuous Growth Mindset.
- Receptivity to Feedback & Code Review Culture. • Proactive Communication & Collaboration.